



# Mechatronic Actuators Trainer

Fundamental Labs – Stepper Motor

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Quanser Consulting Inc. info@quanser.com  
119 Spy Court Phone : 19059403575  
Markham, Ontario Fax : 19059403576  
L3R 5H6, Canada printed in Markham, Ontario.

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电子信息产品污染控制管理办法 (中国 RoHS)



中国客户 Quanser Consulting Inc. 关于关于限制在电子电气设备中使用某些有害成分的指令 (RoHS)。

**CE Compliance** 

This product meets the essential requirements of applicable European Directives as follows:

- 2014/30/EU: Electromagnetic Compatibility Directive (EMC)

**Warning:** This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

# Mechatronic Actuators Trainer – Application Guide

## Stepper Motor Lab

### Why Explore a Stepper Motor?

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Stepper motors are a type of brushless dc motors that rotates in discrete steps. These steps can be achieved by sending electrical pulses to each of the coils. Because the motor moves in precise steps, position and speed can be tracked simply by counting the steps, no feedback sensors are required. Stepper motors can accurately be controlled to get to a specific speed or precisely hold a position while maintaining high torque. Because of how the phases are energized, steppers are also good at maintaining high torque at lower speeds, which can be challenging for other types of motors.

They are ideal for applications that require reliability and repeatability of movement and are easy to operate as an open loop control system. However, the torque of the motor decreases as the speed increase, making them better for lower speed operations.

## Background

This lab is part of the Fundamentals labs for the Mechatronics Actuators Trainer. These labs will guide you through four different types of motors, so you can understand their operating principle and get you comfortable with using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4\_concept\_reviews/),

- 4\_concept\_reviews/Mechatronics/**005\_motor\_stepper**.

## Getting Started

In this lab, you will use the **Actuators Trainer** to learn about the basics of **stepper motors**. The lab will guide you through different ways of driving your stepper motor and the differences between them. You will experiment with them to see the differences in step sizes and speed. You will also be using the datasheet to validate your results.

Before you begin this lab, ensure that the following criteria are met:

- Make sure you have either the provided Stepper motor, or another bipolar stepper motor properly wired to be able to interface with the Actuators Trainer.
- The provided stepper motor is this one. Confirm your motor is wired properly to the 4-pin connector to run it.



- Make sure the Actuators Trainer has been setup and tested. See the [Quick Start Guide \(Quanser/2\\_quick\\_start\\_guides/actuators\\_trainer\)](#) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
  - o If using Python, we recommend using Visual Studio Code.